

Bingxi (Paul) Jiang

(949) 344-0799 | pauljiang817@gmail.com | www.linkedin.com/in/bingxi-jiang | github.com/Bingxi-Jiang

SUMMARY

AI-focused software engineer building agent orchestration systems, multimodal ML pipelines, and production backend services with Python, Node.js, React, Docker, Kubernetes, and cloud infrastructure.

EDUCATION

University of California, Irvine (UCI) | Irvine, CA

Bachelor's Degree, Double Major: Computer Science; Business Information Management | GPA: 3.776/4.0 | Expected June 2028

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript/TypeScript, Java

Frontend: React, Next.js, Vue.js, HTML, CSS

Backend: Node.js, Express, Spring Boot, FastAPI, REST APIs, WebSocket

Databases & Messaging: PostgreSQL, MySQL, MongoDB, Redis, Kafka, Elasticsearch

Cloud & DevOps: GCP, AWS, Docker, Kubernetes, Github Actions, CI/CD, Nginx

AI/ML: PyTorch, Scikit-Learn, Pandas, NumPy, LangChain, LoRA, OpenAI, Gemini, Claude

PROFESSIONAL EXPERIENCE

GTechFin Inc (B2B Fintech startup) | New York, NY

Software Engineer Intern | April 2026 – June 2026

- Built an agentic marketing platform automating TikTok workflows from trend discovery and scripting to AI video rendering.
- Reduced video production cycle by 89%, compressing manual workflow from ~3 hours of editing to ~20 minutes of automated generation.
- Connected Apify, Claude, and HeyGen to discover top videos, extract creative patterns, and generate UGC-style AI marketing videos.
- Engineered a closed-loop feedback system ingesting views, likes, shares, and retention signals to update prompt and creative direction.
- Automated TikTok posting via Content API with caption generation, hashtag packaging, AI-content disclosure, and publish-status polling.

Fresh Road Inc (B2B SaaS startup) | San Jose, CA

Software Engineer Intern | September 2025 – December 2025

- Built an AI-powered B2B auto-callback system using LLM to accelerate inbound lead follow-up and improve conversion readiness.
- Redesigned an event-driven workflow with ownership tracking, reducing coordination and improving operational efficiency by 50%.
- Architected a Node.js orchestration service integrating OpenAI and Gemini providers with routing, fallback, and cost-control logic.
- Built queue-based buffering for async transcription, summarization and CRM sync improving resilience by decoupling provider latency.
- Implemented observability with logs, correlation IDs, latency/error metrics, timeouts, and idempotency safeguards to improve reliability.

Ping An Technology | Guangdong, China

Computer Vision Algorithm Engineer | June 2025 – September 2025

- Designed multi-stage inference pipeline for vehicle damage assessment with multi-view ingestion, prompt reasoning and report generation.
- Productionized VLM training workflow packaging Qwen2.5-VL with LLaMA-Factory achieving 55% recall and 69% precision.
- Built image-quality enhancement module leveraging CLIP semantic signals guiding adaptive processing improving model stability.
- Developed defect detection workflow combining diffusion features with CLIP and PatchCore framing extraction and decision service.
- Engineered production deployment pipeline with Docker containers and Kubernetes orchestration handling 10K+ daily inference requests.

RESEARCH EXPERIENCE

Relational Cognition Lab, UCI | Irvine, CA

Research Assistant | January 2026 – Present

- Built evaluation harness probing representations across Gemma, Qwen, and Llama standardizing prompts, sampling and aggregation.
- Implemented noise-estimation baseline injecting random text into prompts quantifying variance producing defensible comparisons.
- Developed experiment orchestration framework with Python and MLflow tracking runs, metrics and versions ensuring reproducibility.
- Deployed inference service with FastAPI and Redis caching serving embeddings for downstream analysis reducing latency by 70%.

PROJECTS

Noted: Real-Time Multimodal Transcription & Memory System | <https://github.com/Bingxi-Jiang/Noted>

- Architected real-time transcription pipeline streaming 16kHz PCM audio via WebSocket to Deepgram Nova-2, achieving <300ms latency.
- Engineered multimodal memory system combining transcripts, speaker diarization, and Gemini Vision analysis for context preservation.
- Implemented cross-session RAG with embedding similarity search over transcript chunks, enabling context-aware retrieval across folders.
- Established real-time UX with recording, mic mixing, screen analysis and boundary detection, enable zero-interruption multi-hour capture.

Course Helper: Intelligent Academic Parser & ETL Pipeline | www.course-helper.com

- Built schema-driven ETL pipeline converting heterogeneous syllabi into typed JSON reducing manual extraction by 93% per syllabus.
- Added data quality gates with schema validation, anomaly checks and deduplication designing outputs for calendar sync and reminders.
- Deployed processing service with PostgreSQL backend and Redis caching handling 1K+ syllabus uploads daily with automated validation.
- Containerized microservices with Docker and Kubernetes orchestration deploying on AWS EC2 with auto-scaling handling traffic spikes.

FocusTrail: AI Task Breakdown & Real-Time Monitor | <https://github.com/PST-Protocol/FocusTrail>

- Built Gemini-powered task orchestration system with React and Vite to break complex tasks improving execution efficiency by 43%.
- Built Node.js and Express backend with Server-Sent Events streaming real-time focus telemetry from a macOS osascript agent.
- Designed rule-based focus privacy filters detecting distractions while excluding sensitive apps and supporting user-defined policies.
- Deployed full-stack productivity platform on GCP with REST APIs for breakdown, monitor sessions, stats aggregation and calendar history.

AWARDS & LEADERSHIP

- 1st Place, Streamlit AI Hack; built and demoed an AI application using Snowflake Cortex under time-constrained product requirements.
- Event Organizer, Chinese Union in Computer Science; organized technical events and supported student developer community program.